

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – PSEUDOCODE WRITING TASK

Course Code:	CSA0305	Course Title:	Data Structures
CO Assessed:	CO5 – Naturalization (BL4, BL6)	Assessment:	Assessment Tool 1 – Pseudocode Writing Task
Weightage:	35% of CIA	Total Marks:	100
CO5 Statement:	<i>Construct MST using shortest path algorithms and traverse the graph to model and analyse realworld problem.</i>		
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Section / Batch:	2023	Date:	31/08/2026

Code of Conduct

I _____M.Deepika-192372296_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action. Signature of the Student: ____deepika.m_____

Instructions

- This test contains 5 Pseudocode Writing Task questions. Total: 100 Marks.
- When a student answer using pseudocode writing task should evaluate based on the ability to design clear, logical, and efficient pseudocode solutions for data structure problems.
- No negative marking. Unanswered questions carry zero marks.

Section / Topic	Focus	Q Nos	Marks
Minimum Spanning Tree	Kruskal's Algorithm	1	20
Graph Sorting	Topological Sort	2	20
Shortest Path Algorithm	MST & Dijkstra's Algorithm	3	20

Shortest Path Algorithm	Dijkstra's Algorithm	4	20
Minimum Spanning Tree	Prim's or Kruskal's Algorithm	5	20
GRAND TOTAL:		100 Marks	

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	20	Demonstrates complete and clear understanding; handles edge cases effectively	Good understanding; minor gaps in edge case handling	Partial understanding; misses some requirements	Misinterprets problem; major gaps
Code Correctness	30	Fully correct; passes all test cases	Mostly correct; minor errors	Partially correct; several test cases fail	Incorrect or nonfunctional code
Code Quality & Readability	20	Clean, modular, wellcommented, follows standards	Readable with some structure	Limited readability; poor structure	Unorganized and hard to read
Complexity Analysis	20	Accurate time & space complexity with justification	Correct but limited explanation	Incomplete or partially correct	Missing or incorrect
Testing & Validation	10	Thorough testing including edge cases	Adequate testing with few edge cases	Minimal testing	No testing evidence
Faculty In-charge		Course Coordinator		Program Director	
Signature: _____		Signature: _____		Signature: _____	
Date: _____		Date: _____		Date: _____	

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QUESTIONS

Q1

Minimum Spanning Tree -
Kruskal's Algorithm
20 marks

Q2

Graph - Topological Sort
20 marks

Q3

Shortest Path Algorithm -
Minimum Spanning Tree &
Dijkstra's
20 marks

Q4

Shortest Path Algorithm -
Dijkstra's Algorithm
20 marks

Q5

Minimum Spanning Tree -
Prim's or Kruskal's Algorithm
20 marks

Q1 Minimum Spanning Tree - Kruskal's Algorithm

20 marks

Write pseudocode for constructing an MST using Kruskal's Algorithm. A city planning department needs to connect multiple zones with underground fiber-optic cables. Each zone is represented as a vertex, and possible cable connections between zones are represented as edges with installation costs (weights).

Constraints:

- All zones must be connected
- Total installation cost must be minimized
- No redundant connections (no cycles) allowed
- The network must remain fully connected

Your Answer

```
// Step 2: Initialize Disjoint Set (Union-Find)
FOR each vertex v in V
    MAKE-SET(v) // Each zone starts as its own separate component
END FOR

edgeCount = 0
```

Save Answer

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QUESTIONS

Q1

Minimum Spanning Tree -
Kruskal's Algorithm
20 marks

Q2

Graph - Topological Sort
20 marks

Q3

Shortest Path Algorithm -
Minimum Spanning Tree &
Dijkstra's
20 marks

Q4

Shortest Path Algorithm -
Dijkstra's Algorithm
20 marks

Q5

Minimum Spanning Tree -
Prim's or Kruskal's Algorithm
20 marks

Q2 Graph - Topological Sort

20 marks

A DAG represents task dependencies. Write pseudocode for Topological Sort using DFS with cycle detection.

Constraints:

- Detect cycle before sorting
- Use DFS approach

Your Answer

```
for each vertex v in G:
    color[v] = WHITE
Stack = empty stack
hasCycle = false
for each vertex v in G:
    if color[v] == WHITE:
```

Save Answer

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QUESTIONS

Q1 ✓
Minimum Spanning Tree -
Kruskal's Algorithm
20 marks**Q2** ✓
Graph - Topological Sort
20 marks**Q3** ✓
Shortest Path Algorithm -
Minimum Spanning Tree &
Dijkstra's
20 marks**Q4** ✓
Shortest Path Algorithm -
Dijkstra's Algorithm
20 marks**Q5** ✓
Minimum Spanning Tree -
Prim's or Kruskal's Algorithm
20 marks**Q3 Shortest Path Algorithm - Minimum Spanning Tree & Dijkstra's**

20 marks

Write pseudocode integrating MST + Dijkstra for Smart transportation system.

Constraints:

- a. Use MST for infrastructure
- b. Use shortest path for routing

Your Answer

```
ALGORITHM Build_Transportation_Infrastructure(Graph G)
Initialise empty set MST_Edges.
Initialise Union-Find structure for all nodes (locations).
Sort all edges of G in ascending order by weight (cost).
```

```
FOR each edge (u, v, weight) in sorted edges:
```

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QUESTIONS

Q1 ✓
Minimum Spanning Tree -
Kruskal's Algorithm
20 marks**Q2** ✓
Graph - Topological Sort
20 marks**Q3** ✓
Shortest Path Algorithm -
Minimum Spanning Tree &
Dijkstra's
20 marks**Q4** ✓
Shortest Path Algorithm -
Dijkstra's Algorithm
20 marks**Q5** ✓
Minimum Spanning Tree -
Prim's or Kruskal's Algorithm
20 marks**Q4 Shortest Path Algorithm - Dijkstra's Algorithm**

20 marks

Shortest Path with "Toll-Free" Voucher: You have a voucher that allows you to traverse one edge for free (weight = 0). Write pseudocode to find the shortest path from S to D utilizing this voucher optimally.

Your Answer

```
ALGORITHM DijkstraWithVoucher (Graph G, Source S, Destination D):
```

```
// dist[v][0] = shortest distance to v without using voucher
// dist[v][1] = shortest distance to v after using voucher
for each vertex v in G:
  dist[v][0] = INFINITY
```

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QUESTIONS

Q1 ✓
Minimum Spanning Tree -
Kruskal's Algorithm
20 marks

Q2 ✓
Graph - Topological Sort
20 marks

Q3 ✓
Shortest Path Algorithm -
Minimum Spanning Tree &
Dijkstra's
20 marks

Q4 ✓
Shortest Path Algorithm -
Dijkstra's Algorithm
20 marks

Q5 ✓
Minimum Spanning Tree -
Prim's or Kruskal's Algorithm
20 marks

Q5 Minimum Spanning Tree - Prim's or Kruskal's Algorithm

20 marks

Second Best Spanning Tree: Design pseudocode to find the "Second Best" MST.

Hint: Find the MST first, then for every edge in the MST, try removing it and finding the next best edge to replace it.

Your Answer

Remove edge (u, v) from MST_edges temporarily
Perform DFS/BFS from u to find all reachable vertices → set A
The remaining vertices form set B
Search through all edges in G:
For each edge (x, y) not in MST_edges:
If (x in A and y in B) or (x in B and y in A):

[Save Answer](#)